

# Introduction to RAG

Retrieval-Augmented Generation

Pre-Read Material for Students

Scaler Academy — AI & ML Track

Hello! This document is your **quick starter guide** before the live session on RAG. It is written in simple language — no PhD required. Read this once before class and you will be fully ready to follow along.

| Reading Time | Pages   | Level    | Track   |
|--------------|---------|----------|---------|
| ~15 minutes  | 3 pages | Beginner | AI & ML |

## What's inside this document:

1. Section 1 — What is RAG and why do we need it?
2. Section 2 — How basic RAG works (step by step)
3. Section 3 — What should you know before the class (Prerequisites)
4. Section 4 — What you will learn in the session

## Section 1 — What is RAG and Why Do We Need It?

Let's start with a simple everyday analogy.

Imagine you are asking a very smart friend a question about something that happened last week. But this friend has been living in a bunker since 2023 — no internet, no news, no updates. He is extremely intelligent, but he only knows what he knew before going underground. If you ask him about yesterday's cricket score, he will either say *I don't know* or — even worse — *make something up* and sound very confident about it.

**That friend is an LLM (like ChatGPT or Claude).** It is brilliant but it has a knowledge cutoff. It does not know what happened after it was trained. And it cannot look at your company's private files, your own notes, or any document you haven't given it.

RAG — Retrieval-Augmented Generation — solves this by giving the LLM the ability to **look things up first**, then answer. Think of it like an open-book exam instead of a closed-book one. The AI is still doing the reasoning, but now it can read the right pages of the book before answering you.

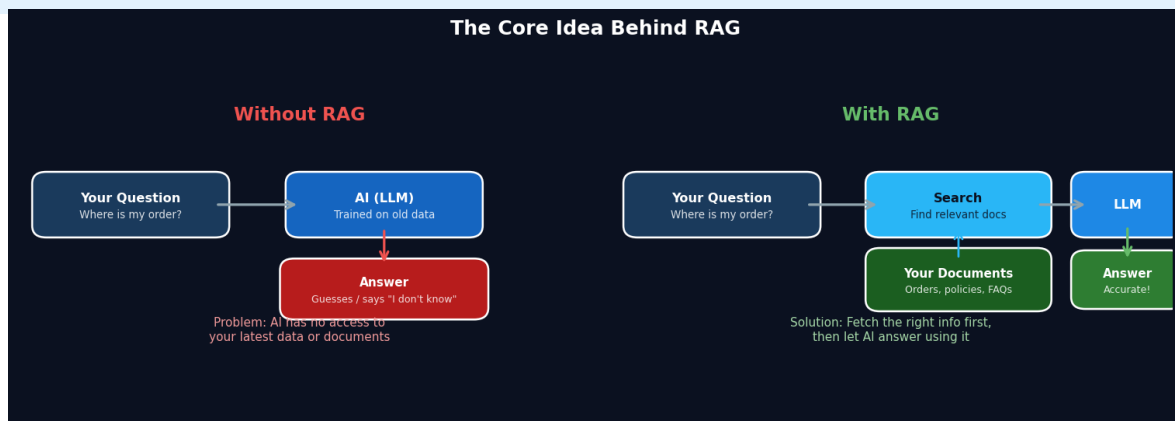


Figure 1: Without RAG the AI guesses from old memory. With RAG it finds the right information first, then answers.

### The Three Problems RAG Fixes

**Old Knowledge:** LLMs stop learning at a point in time. Your business data from last month? Unknown to the model. RAG lets you plug in fresh documents any time.

**Making Things Up (Hallucination):** When an LLM doesn't know something, it does not stay quiet. It generates a confident-sounding wrong answer. RAG grounds the answer in real documents, so the model has something true to refer to.

**Private Data:** The model was never trained on your company's internal policies, your customer data, or your product manuals. RAG lets you connect these private sources safely without retraining the entire model.

## Section 2 — How Basic RAG Works

Basic RAG has two phases. Phase 1 prepares your documents (you do this once). Phase 2 handles each user question (this runs every time someone asks something).

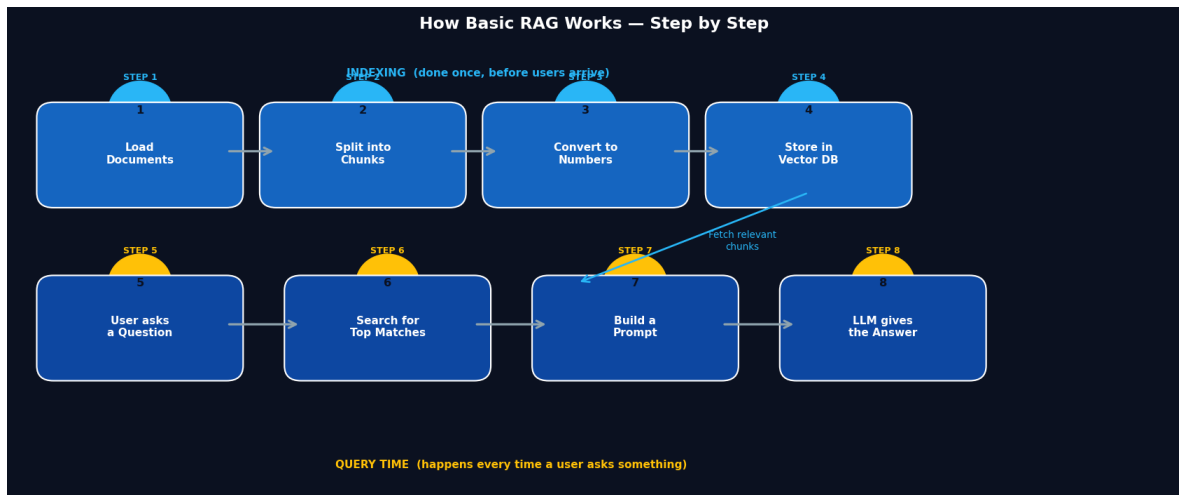


Figure 2: The 8 steps of basic RAG — 4 steps to prepare documents, 4 steps to answer a question.

## Let's Understand Each Step in Plain English

| Step               | What Happens                                                         | Simple Example                                                       |
|--------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>1. Load</b>     | Read your documents into the system                                  | Upload your 50-page product manual as a PDF                          |
| <b>2. Chunk</b>    | Cut long documents into small pieces                                 | Break the 50-page PDF into 200 small paragraphs                      |
| <b>3. Embed</b>    | Convert each chunk into a list of numbers that captures its meaning  | Paragraph about "returns policy" becomes a unique number fingerprint |
| <b>4. Store</b>    | Save all these number fingerprints in a special database (Vector DB) | Think of it like a library catalogue but for meaning, not titles     |
| <b>5. Question</b> | User types a question                                                | "What is the return policy for electronics?"                         |
| <b>6. Search</b>   | Find the chunks most relevant to the question                        | System finds the 3 paragraphs about return policy                    |
| <b>7. Prompt</b>   | Combine: question + relevant chunks -> send to LLM                   | "Here is the return policy text... Now answer this: ..."             |
| <b>8. Answer</b>   | LLM reads the context and gives a grounded answer                    | "You can return electronics within 30 days with receipt."            |

## Section 3 — What You Need to Know (Prerequisites)

You do not need to be an expert. Here is an honest breakdown of what will help you get the most out of this session. We will quickly recap anything marked Good to Have at the start of class.

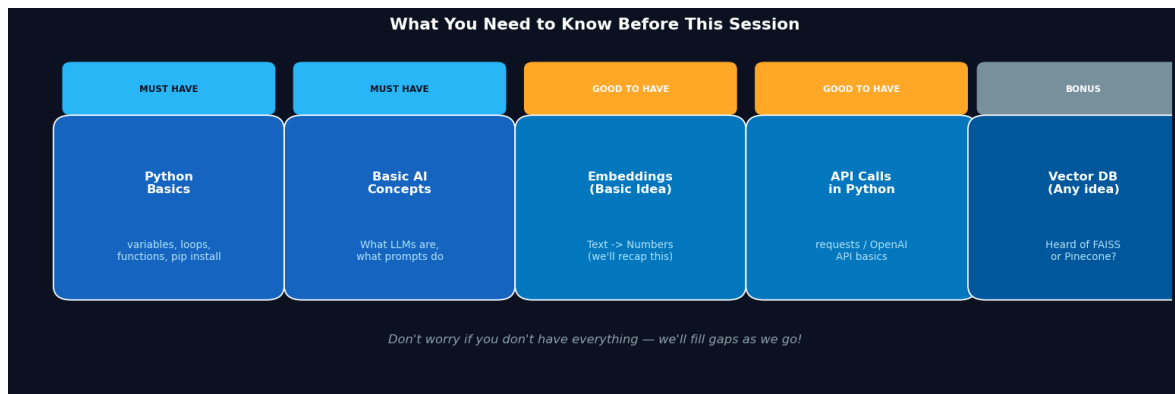


Figure 3: Prerequisites for this RAG session. Blue = you need this. Orange = helpful. Grey = bonus.

| Topic                   | What Exactly You Need                                                                              | Must Have?   |
|-------------------------|----------------------------------------------------------------------------------------------------|--------------|
| Python Basics           | Write a simple function, install packages with pip, read a file, use a loop and conditionals       | YES          |
| What is an LLM          | Know that models like ChatGPT are trained on text and generate responses based on that training    | YES          |
| What is a Prompt        | Understand that the text you send to an AI is called a prompt and it affects the response          | YES          |
| Embeddings (rough idea) | Text can be converted to numbers. Similar text has similar numbers. We will explain this in depth. | Good to Have |
| API Calls in Python     | Made at least one API call using requests or the OpenAI library. Not mandatory but helpful.        | Good to Have |
| What is a Database      | Basic idea that databases store and retrieve data. No SQL required.                                | Good to Have |

Not sure if you are ready? Ask yourself: Can I write a Python function that opens a text file and prints each line? If yes — you are good to go!

## Section 4 — What You Will Learn in the Session

By the end of the live class you will be able to:

- + Explain what RAG is and why it exists — to anyone, in simple words
- + Build a working RAG pipeline from scratch using Python and LangChain
- + Load your own documents, chunk them, embed them, and store them in a vector database
- + Connect this to an LLM so it can answer questions using your documents
- + Understand where basic RAG can go wrong and what to watch out for

Tools we will use: Python, LangChain, OpenAI API, FAISS (or Chroma), Google Colab (no local setup needed). Everything will be provided in the session.